

Conditioning as Group Action: Exact Compositional Conditioning at FiLM Cost

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Abstract

Networks are conditioned by scaling features (FiLM) or by generating weights (hypernetworks). Both learn what each condition does. Neither learns what two conditions do together unless training data shows them together. We condition by rotating features, with rotation angles a linear function of the condition. Rotations add, so combined conditions compose automatically, and the network handles combinations it never saw. **CGA** (Conditioning as Group Action) formalizes this: the condition enters the operator linearly in an abelian Lie algebra, $T(c) = P \exp(A(Wc))P^\top$, so $T(c_1+c_2) = T(c_1)T(c_2)$ holds for any weights. A diagonal algebra yields a compositional FiLM variant; the rotation algebra with scalar position is RoPE. Under a pre-registered protocol (margins, tests, and splits committed before every run; conditioning cost within $1.2\times$ FiLM), CGA posts the lowest error on never-trained condition combinations across five transformation suites, with complete seed separation ($p=9.1\times 10^{-5}$, $N=10$) and zero-shot extrapolation to combinations longer than any combination seen in training. Swapping its linear map for an MLP raises error by up to $4.3\times 10^4\times$; the most expressive baselines generalize worst. Three registered gates returned negatives that bound the method: a fit penalty on rich scenes, a content-conditioning failure (rotations move information and cannot add it), and multi-step rollouts, where every conditioner compounds error equally unless a latent-consistency loss is added. The negatives specify the successor. **Complex FiLM**, magnitude for content and phase for composition, beats FiLM by 36% on unseen combinations and matches it on content ($0.97\times$) under two further pre-registered gates. The same structure yields guidance without classifier-free guidance (CFG) and its second forward pass: raising the operator to a power distorts $2\text{--}5\times$ less than CFG at every strength. One operator, both roles, FiLM cost.

1 Introduction

Nearly every conditional model must inject something into its computation: a class, an action, a timestep, a text embedding. Call this conditioning. Diffusion models modulate every block on the timestep and class via adaLN [2]; visual reasoning, speech, and RL systems modulate features on questions, speakers, and goals [1]; world models condition latent dynamics on actions. Almost all of this conditioning is carried by three mechanisms: concatenation, feature-wise affine modulation (FiLM), or a hypernetwork that emits weights from the condition.

Two of these bracket an expressivity axis; we set concatenation aside, and carry it as a baseline arm. FiLM’s diagonal transform is cheap and stable but cannot mix features. Hypernetworks can express any linear operator but pay quadratic cost and, as we show, memorize the training conditions: their error on unseen combinations of familiar conditions is worse than FiLM’s, despite $43\times$ the conditioning compute. The middle of this axis, structured operator families between diagonal and dense, is well explored for weight fine-tuning [6–9] and largely unexplored in FiLM’s role: per-sample, input-conditioned modulation of activations.

Conditioning plays two roles, and the distinction runs through this paper. A condition can name a *change* to apply, such as a pose offset or an edit, or it can name the *content* to produce, such as a class or a caption. FiLM serves both. We ask which structure, at FiLM-level cost, buys compositional generalization: correct behavior on combinations of conditions never seen together in training. Our answer imports a mechanism from positional encoding. RoPE [4] and its generalization GRAPE [5] encode position n as a group action $G(n) = \exp(n\omega L)$; because position enters linearly in the Lie algebra, $G(n+m) = G(n)G(m)$ holds exactly. We transplant this construction from scalar positions in attention to arbitrary condition vectors in activation modulation.

Contributions. (1) **Framework.** CGA treats conditioning as a group representation of the condition space; FiLM, RoPE, and GRAPE drop out as special cases, and a two-line proposition makes condition composition exact for any

weights (§3). **(2) Protocol.** A pre-registered, budget-fair evaluation: margins, statistical tests, and single-read splits committed to version control before every run, with a shared counter enforcing parameter and FLOP parity; we report every verdict, including negatives against our own predictions (§4). **(3) Where it works, and where it stops.** CGA has the lowest error on unseen condition combinations in all five suites where the condition is a transformation (S1, S2, S3, S3b, S4), by up to five orders of magnitude, and in every image suite the more expressive the conditioner, the worse it generalizes. Three of the eight suites returned pre-registered negatives, and each names a setting where the composition law of Proposition 1 buys nothing: S4, which CGA wins on unseen combinations but loses on training fit; S5, content conditioning; and S6, multi-step rollouts (§5). **(4) Successor and guidance.** The negatives specify Complex FiLM, validated under two further gates as a FiLM replacement covering both conditioning roles, whose phase channel also yields guidance as an exact operator power: 2–5× less distortion than CFG with no unconditional pass (§5).

2 Related work

FiLM and conditional normalization. FiLM [1] and its normalized variants (conditional BN/LN, adaLN in DiT [2]) apply per-feature affine transforms generated from a condition. All are diagonal. We keep FiLM’s role and enrich only the operator.

Structured operators in parameter-efficient fine-tuning (PEFT). Low-rank, orthogonal, and butterfly operator families are well studied as weight fine-tuning: LoRA [6], OFT [7], BOFT [8], HRA [9], Group-and-Shuffle [10]; He et al. [11] give a unified view. These learn one fixed transform per task. Our setting differs on every axis that matters: per-sample, input-conditioned, activation-space, and evaluated for compositional generalization. Hypernetworks [3] generate weights from conditions and inherit the memorization behavior we quantify.

Position encodings as group actions. RoPE [4] rotates query-key pairs by angles linear in position; GRAPE [5] generalizes this to group representations with exact relative laws. The mechanism is theirs; our contribution is the transplant to arbitrary condition vectors in FiLM’s role, with a learned basis, and the budget-fair compositional evidence.

Compositional generation. Montero et al. [12] showed reconstruction-trained generative models fail on unseen factor combinations. Score and energy composition [23] composes concepts across separate conditional passes; we compose conditions inside the operator, in one pass. Symmetry-based representation learning [13, 14] argues latents should carry group actions; we supply the conditioning-side mechanism.

Nearest prior work. We searched deliberately for work that would invalidate this paper. The closest hits: Feature-space rotations with exact composition exist for geometric transformations [17]; abelian exponential-map operators for disentanglement [22]; attribute operators for recognition [26]; multilinear attribute mixing for unseen combinations [27]; per-sample amplitude-and-phase activation modulation in periodic implicit networks [20]; and composition by complex phase addition in knowledge-graph embeddings [21]. Rotation-structured latent world models with latent-prediction losses originate with Quessard et al. [15], Caselles-Dupré et al. [16], and the homomorphism autoencoder [18], whose composition property is emergent rather than imposed; latent consistency objectives are standard in model-based RL [19]. Guidance has been moved into the conditioning path by distilling a CFG teacher [25], and CFG’s high-scale distortion is established [24]. We claim none of these components. We claim the combination: a group-structured operator in FiLM’s role at budget parity, composition imposed exactly by a linear-in-algebra map, magnitude split from phase as content and composition channels, guidance as an exact power, and verdicts from pre-registered gates.

3 The CGA framework

Let $x \in \mathbb{R}^d$ be features and $c \in \mathbb{R}^k$ a condition. FiLM computes $y = \gamma(c) \odot x + \beta(c)$; hypernetworks emit $W(c)$. All are instances of

$$y = T(c)x + \beta(c), \quad (1)$$

and they differ only in where the per-sample operator $T(c)$ comes from.

Definition 1 (Compositional conditioner). $T : \mathbb{R}^k \rightarrow \text{GL}(d)$ is compositional if it is a homomorphism from $(\mathbb{R}^k, +)$: $T(c_1+c_2) = T(c_1)T(c_2)$ and $T(0) = I$.

Proposition 1 (Exact composition by construction). Let $\mathfrak{a}$ be an abelian subalgebra of $\mathfrak{gl}(d)$, $A : \mathbb{R}^m \rightarrow \mathfrak{a}$ linear, $W \in \mathbb{R}^{m \times k}$, and P orthogonal. Then $T(c) = P \exp(A(Wc)) P^\top$ is a compositional conditioner for every W, P . If $\mathfrak{a}$ is skew-symmetric, $T(c)$ is orthogonal.

Proof. $\exp(X)\exp(Y) = \exp(X+Y)$ for commuting X, Y ; $A \circ W$ is linear; conjugation preserves products; $\exp(0) = I$. $\square$

Composition is usually a behavior the network must learn from data. Here it is a property of the parametrization. It holds at initialization, during training, and far outside the training distribution. Training only decides which directions the condition rotates and how fast.

Special cases. Choices of $(\mathfrak{a}, c, P)$ recover deployed mechanisms:

$\mathfrak{a}$	condition	P	recovered mechanism
$\text{diag}(\mathbb{R}^d)$	c	I	<i>exponential FiLM</i> : $y = e^{Wc} \odot x$ (compositional)
$\mathfrak{so}(2)^{d/2}$	$c = n$ (scalar position)	I	RoPE [4]
$\mathfrak{so}(2)^{d/2}$	$c = n$, learned planes	learned	multiplicative GRAPE [5]
$\mathfrak{so}(2)^{d/2}$	general $c \in \mathbb{R}^k$	structured	this work (FiLM’s role)
$\text{diag} \times \mathfrak{so}(2)^{d/2}$	general c	canonical	Complex FiLM (§5)

Standard FiLM reaches the same diagonal operators through an MLP head $\gamma(c)$. The operators match; only the composition law is missing, and §5 shows that its absence is what breaks FiLM on unseen combinations.

Cost. With $\mathfrak{a} = \mathfrak{so}(2)^{d/2}$, $\exp$ is closed-form planar rotation: $3d$ FLOPs, no matrix exponential. The basis P is the whole budget question. A dense learned P costs $4d^2$ FLOPs per sample, which is $1.52\times$ FiLM’s entire conditioning path at $d=128$. Two fixes work. A structured orthogonal P (five block-diagonal Cayley layers with fixed shuffles, following Group-and-Shuffle) brings the total to $1.11\times$ FiLM, with 4,800 parameters against the 8,128 dimensions of $\text{SO}(128)$; that is enough, because P only needs the planes the condition acts on. Inside a transformer, no explicit P is needed: the adjacent dense projections can absorb P , exactly as RoPE relies on the query and key projections rather than on a basis of its own. Linearity also gives the operator a strength dial for free: $T(c)^w = T(wc)$, so raising the operator to a power is the same as scaling the condition, and §5 uses this in place of CFG’s extrapolation between a conditional and an unconditional prediction. The operator equals the identity at initialization and at $c=0$, and every singular value is 1. Unit tests pin all three properties. Complex FiLM extends the algebra to $\text{diag} \times \mathfrak{so}(2)^{d/2}$: each feature pair is multiplied by $m e^{i\theta}$, with magnitude from FiLM’s expressive head (the content channel) and phase linear in the condition (the composition channel), at $0.84\times$ FiLM’s cost in our harnesses.

4 Pre-registered, budget-fair protocol

Vocabulary. An *arm* is one conditioning method, trained under the shared recipe. A *suite* is one experiment: a fixed backbone, dataset, and training loop that all of its arms share. A *gate* is that suite’s pass-or-fail rule, written down before the run as a conjunction of *criteria*: typically a margin over the strongest baseline on unseen combinations, a no-regression criterion on the training distribution, and a budget criterion. A suite passes only if all of its criteria pass, and we report the outcome either way.

Design. Every suite is pre-registered: success margins, statistical tests ($N=10$ seeds, one-sided Mann–Whitney $p \leq 0.01$, Cliff’s $|\delta| \geq 0.474$), and data splits were fixed and committed to version control before each run. Each test split is read once, after all selection on a validation split. All arms share one backbone and one training recipe. A shared counter enforces the budget: our conditioning path stays within $1.20\times$ FiLM’s FLOPs and $1.05\times$ the smallest hypernetwork baseline’s parameters; baselines may be larger, and reach $48\times$ our parameter count. Deviations are dated amendments in the committed specifications, plus one accounting erratum that our own audit caught and that downgraded a favorable verdict.

Table 1: Error on never-trained condition combinations: mean (seed sd) over 10 seeds. CGA is lowest in every column, with complete seed separation against the strongest hypernetwork baseline ($p=9.1\times 10^{-5}$, $\delta=-1.0$). Throughout, OOD means out-of-distribution: a condition combination never seen in training. S1–S3b pass their full pre-registered gate; S4 fails on a separate training-fit criterion (text). S7 reruns the S3 harness with Complex FiLM. **Columns are not comparable to one another:** S1 and S2 report operator MSE, S3–S7 report per-pixel MSE. **Suites:** S1 synthetic, aligned; S2 synthetic, hidden basis; S3 dSprites; S3b dSprites plus a categorical shape factor; S4 3D Shapes; S5 content conditioning in diffusion; S6 world-model rollouts; S6' the same with a latent-consistency loss; S7 Complex FiLM; S8 guidance.

Conditioner	S1	S2	S3	S3b	S4	S7
FiLM	$1.3 (0.1)\times 10^{-2}$	$1.6 (0.04)\times 10^{-2}$	$3.0 (0.2)\times 10^{-3}$	$5.9 (0.2)\times 10^{-3}$	$4.5 (1.0)\times 10^{-3}$	$3.1 (0.2)\times 10^{-3}$
Concat-MLP	$1.5 (0.3)\times 10^{-2}$	$1.1 (0.4)\times 10^{-2}$	$2.9 (0.1)\times 10^{-3}$	$6.3 (0.2)\times 10^{-3}$	$5.4 (5.2)\times 10^{-3}$	–
Cond-LN	$2.4 (0.1)\times 10^{-2}$	$2.7 (0.05)\times 10^{-2}$	$3.2 (0.2)\times 10^{-3}$	$6.1 (0.3)\times 10^{-3}$	$8.6 (13.1)\times 10^{-3}$	–
Hypernetwork	$1.9 (0.3)\times 10^{-3}$	$3.4 (0.9)\times 10^{-3}$	$3.8 (0.3)\times 10^{-3}$	$7.0 (0.4)\times 10^{-3}$	$9.3 (2.3)\times 10^{-3}$	$3.8 (0.2)\times 10^{-3}$
Dyn. low-rank	$5.7 (0.5)\times 10^{-3}$	$6.1 (0.4)\times 10^{-3}$	$5.1 (0.9)\times 10^{-3}$	$8.1 (0.4)\times 10^{-3}$	$2.0 (0.4)\times 10^{-2}$	–
MLP-head (abl.)	–	$1.1 (0.2)\times 10^{-3}$	$2.7 (0.2)\times 10^{-3}$	$5.0 (0.2)\times 10^{-3}$	$3.3 (0.8)\times 10^{-3}$	–
Complex FiLM (linear mag.)	–	–	–	–	–	$2.0 (0.1)\times 10^{-3}$
Complex FiLM	–	–	–	–	–	$2.0 (0.2)\times 10^{-3}$
CGA (ours)	$7.4 (2.0)\times 10^{-4}$	$3.3 (2.2)\times 10^{-8}$	$1.8 (0.1)\times 10^{-3}$	$4.0 (0.1)\times 10^{-3}$	$1.2 (0.2)\times 10^{-3}$	$1.8 (0.1)\times 10^{-3}$

Threats to validity, and the design answer. The benchmarks are self-constructed, so we built them to remove judgment from scoring: dSprites and 3D Shapes are deterministic factor-to-image maps, so every held-out condition has a unique ground-truth image and pixel error is an objective compositional metric, with no perceptual score and no human rating. Pre-registration removes outcome-contingent analysis; the verdicts below include three negatives, one of which falsified our own headline prediction. Every number in this paper regenerates from the committed logs with one command per suite; re-running a suite from scratch costs one training run per arm per seed.

Arms. FiLM; conditional LayerNorm; Concat-MLP; a dense hypernetwork $I+\Delta W(c)$; a dynamic low-rank map $I+A(c)B(c)^\top$; the Lie operator; Complex FiLM (both variants); and an MLP-head ablation of our own method (same basis, same operator family, more compute, no linearity).

Suites. **S1/S2 (synthetic):** learn $y = M(c)x$, where c selects a combination of 8 rotation primitives; S2 hides the primitives behind a random basis and tests all 56 never-trained *triples*. **S3/S3b (dSprites) and S4 (3D Shapes):** image transformation: encode a real image, apply $T(\Delta)$ on the learned latent for a factor change Δ , decode, and score pixel error against the true transformed image; OOD means never-trained *types* of two-factor changes; S3b/S4 add a categorical shape factor. **S5 (content):** a small diffusion transformer generates dSprites images from full factor specifications. **S6 (rollouts):** a world model applies the conditioning arm recurrently, one action per step; horizons 10 and 20 after training on 3. **S7 (successor):** Complex FiLM through the S3 and S5 harnesses. **S8 (guidance):** condition-dropout training, then CFG at strength w for FiLM against condition powering for Complex FiLM, scored against deterministic ground truth per strength.

5 Results

(1) When conditions form a group, composition is exact to numerical precision. On S2, CGA reaches 3.3×10^{-8} error on unseen pairs and 5.3×10^{-8} on never-trained triples; the best baseline sits five orders of magnitude higher and every baseline degrades as combinations lengthen. Learned angles match the true generators to 2×10^{-4} rad: reading the weights tells you which plane each condition rotates, and how fast (Figure 3).

(2) The advantage survives learned representations. On dSprites every arm fits the training distribution equally well, isolating composition. On S3, CGA reduces error on unseen combinations by 53.8% against the strongest baseline, at 39× fewer conditioning FLOPs; on S3b, which adds a categorical shape factor, by 42.9%. The same result in a second

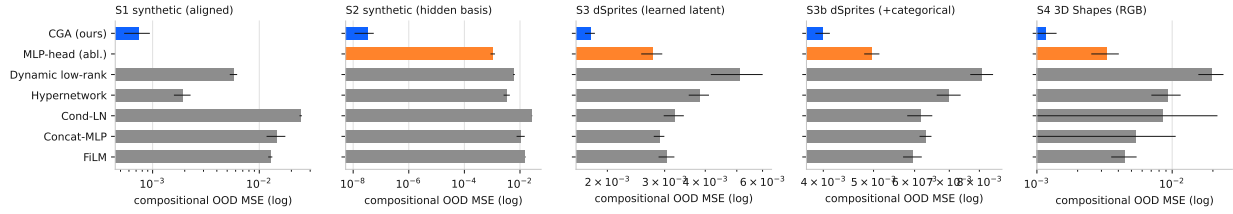


Figure 1: Error on unseen condition combinations by conditioner (log scale; blue = ours, orange = MLP-head ablation), across the five transformation suites.

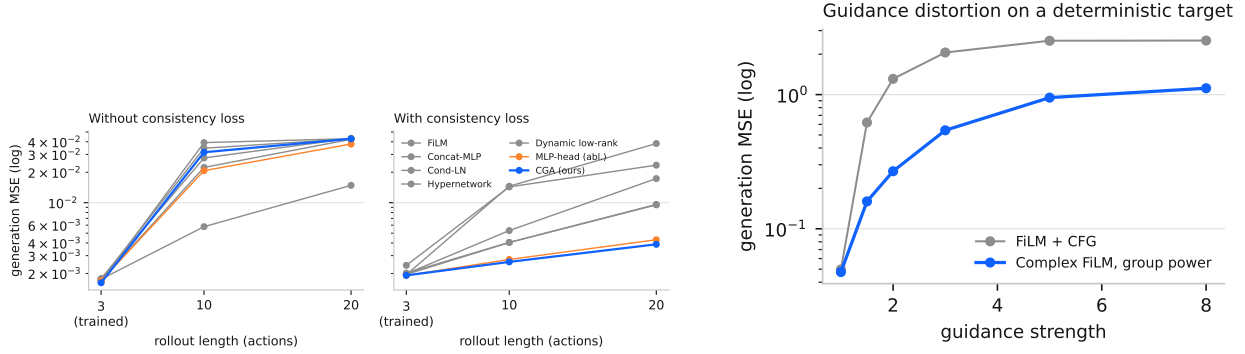


Figure 2: **Left (S6/S6')**: rollout error versus horizon, without and with the latent-consistency loss; with it, the isometric operator stays flat where every baseline drifts. **Right (S8)**: generation error versus guidance strength; CFG's linear extrapolation distorts 50.7 $\times$ from strength 1 to 8, condition powering 23.7 $\times$, staying 2–5 $\times$ lower throughout.

form: going from trained to unseen combinations multiplies CGA's error by 1.25 $\times$ (S3) and 1.63 $\times$ (S3b), and every baseline's by 2.1–3.3 $\times$. We call that multiplier the *compositional gap*.

(3) The linear map is the mechanism. The ablation keeps the basis and the operator family, spends more compute, and swaps only the linear map for an MLP. It loses by $4.3 \times 10^4 \times$ on synthetic triples and 1.55 $\times$ on dSprites.

(4) More capacity makes it worse. The two hypernetwork arms can express any operator. They post the largest compositional gaps of any arm on every image suite: 2.7–3.3 $\times$ on dSprites and 13–26 $\times$ on 3D Shapes, against CGA's 1.13–1.63 $\times$. Their capacity memorizes the training combinations.

(5) Three negatives, scored by the criteria we registered in advance. On 3D Shapes, CGA has the lowest error on unseen combinations (87.5% below the hypernetwork) but its training error is 1.46 $\times$ the hypernetwork's, which trips the pre-registered no-regression criterion. In the content suite (S5), rotation conditioning fails: pixel MSE 0.31 against FiLM's 0.04, an 8 $\times$ regression and the ablation fails identically: a rotation moves information and cannot add any, so it cannot carry “generate this image.” In the rollout suite, every arm compounds error at the same rate; each step adds a small mismatch between the predicted and the true latent. A contracting operator shrinks past mismatches; an isometry preserves them, so they accumulate.

(6) A consistency loss completes the rollout recipe. Training with $\lambda \|T(a)z_s - z_{s+1}\|^2$, applied to every arm equally, inverts the picture: the rotation operator's horizon-20 error goes flat at 0.0039 (growth 2.0 $\times$) against the hypernetwork's 0.038 (growth 19.8 $\times$) and FiLM's 0.017 (Figure 2, left). The loss is standard in model-based RL [19], and rotation transitions with latent-prediction objectives appear in the homomorphism autoencoder [18]; our contribution is to separate the two effects: adding the same loss to every arm at the same budget shows that neither the loss nor the isometric operator fixes rollouts alone, and that they work only together. A null result sharpens this. We tried the obvious alternative and it failed: shrinking the latent by a fixed factor at every step, swept from 0.003 to 0.1, leaves

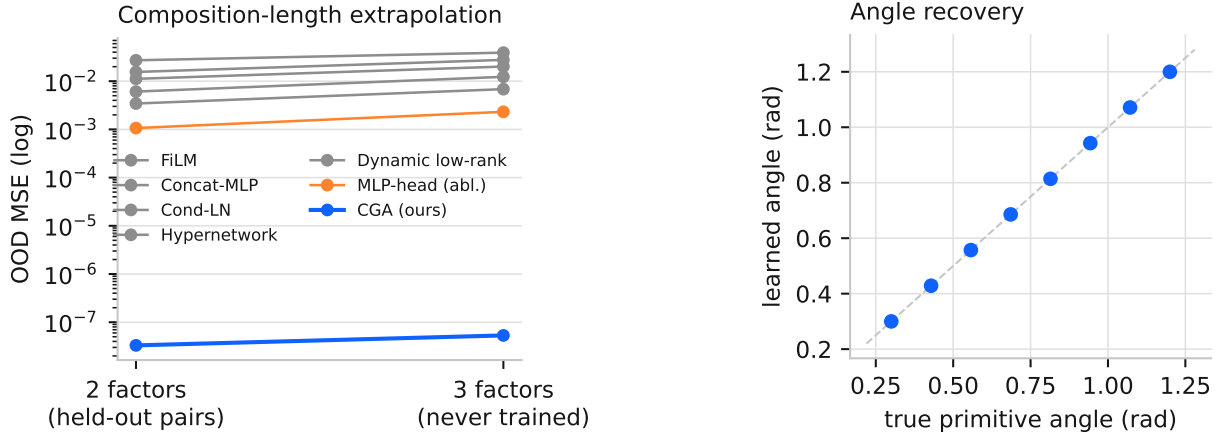


Figure 3: **Left (S2):** extrapolation to composition length 3, never trained; ours is flat at 10^{-8} and every baseline degrades. **Right (S1):** learned singleton angles recover the true generators to 2×10^{-4} rad.

horizon-20 error unchanged. Whatever removes the accumulated mismatch has to depend on the state rather than be a constant. A separate single-step criterion fails, so S6 counts as a negative in Table 1 even though its horizon-20 result is favorable.

(7) The negatives specify the fix, and it works. Complex FiLM multiplies feature pairs by $m e^{i\theta}$. Under two pre-registered gates it beats FiLM by 36% on unseen change types at fit parity and $0.84\times$ FiLM’s cost, and matches FiLM on content generation at $0.97\times$ its error (pre-registered non-inferiority margin $1.10\times$), where the pure rotation failed by $7\times$. Giving the magnitude channel a linear map instead of an MLP fails on that same content metric (0.19 against FiLM’s 0.04): content needs the expressive head. FiLM is the magnitude channel on its own; the full complex multiplication does both jobs.

(8) Guidance without the second pass. CFG’s distortion at high scales is established [24]; our benchmark isolates it against an exact alternative, because the conditioning target is deterministic. CFG’s error grows $50.7\times$ from strength 1 to 8; powering the condition through the phase channel grows $23.7\times$ ($p=1.6\times 10^{-4}$), stays $2\text{--}5\times$ lower at every strength above 1, matches at native strength ($0.96\times$), and needs no unconditional pass and no distillation [25] (Figure 2, right). Neither method improves over strength 1 here: the target is deterministic, so there is no distribution to sharpen, exactly as the suite’s registered caveat stated.

6 Practical guidance

Three rules for practitioners, from the eight suites. **Use phase conditioning when your conditions are changes that combine.** Edit deltas, attribute offsets, pose or camera adjustments, applied once. Cost matches FiLM, fit matches, and error on never-trained combinations is $1.8\text{--}8\times$ lower on real images ($42.9\text{--}87.5\%$ reduction against the strongest baseline) and up to $10^5\times$ lower when the conditions form a group exactly. FiLM gives no warning when it fails this way: its training loss looks normal, and only the unseen combinations are wrong. CGA’s composition law rules that failure out by construction. **For conditions that say what to generate, use Complex FiLM or keep FiLM.** Pure rotation fails there ($7\times$); Complex FiLM matches FiLM on content ($0.97\times$) while keeping the composition wins, so it is the candidate drop-in at our evidence scale. **In rollouts, pair an isometric operator with a consistency loss.** Without the loss, all conditioners compound error at the same rate; with it, the rotation operator stays flat where FiLM grows $4.5\times$ and a hypernetwork $10\times$.

7 Limitations

All evidence comes from controlled 64×64 benchmarks with known factors, width-128 models, and a single conditioning site; nothing here is validated at production scale, and the natural deployment experiment is the adaLN site of a text-conditioned diffusion transformer. Learned latents do not support exact group action, so the advantage on real images is the 1.8–8× error reduction quoted above, not the $10^5\times$ of the synthetic suites. S4 shows orthogonality can cost in-distribution fit on rich scenes. The guidance benchmark measures distortion only, and cannot measure the distribution-sharpening benefit CFG provides when the target is multimodal. Conditions far from any group action, such as open-ended text, remain untested; the learned-condition-geometry extension is future work.

Reproducibility statement. We release the code, the pre-registrations, and the raw logs. Every suite has a specification committed before its decision run; a registry maps each suite label in this paper to the module that produced it, the log it wrote, and its verdict, so a replicator reproduces any row with a single command. Every number, table, and figure here regenerates from those append-only logs. Unit tests pin the operator invariants (identity at initialization and at $c = 0$, orthogonality, exact composition), the budget counters, and the registry itself. All amendments and one accounting erratum, a FLOP undercount our own audit caught that downgraded a favorable verdict, are disclosed with the results.

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